The role of data networking in the management of BT's SDH network

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This paper outlines the history behind the current realisation of the management data communications network for synchronous digital hierarchy (SDH), explaining the various international standards inputs, supplier implementations and illustrating the architecture employed. Message addressing methods are discussed together with the protocols over which they are used. The dimensioning of this network is a key issue of ongoing discussion and impacts the provisioning rates and response times for SDH.

1. Introduction

Data networks within BT are used for many purposes. This paper describes one such use — to control/ manage BT's synchronous digital hierarchy (SDH) network. It gives an overview of the SDH network architecture including the management architecture. It then describes the management communications implementation and the router network which interconnects it.

1.1 SDH background

One of the main benefits of the SDH over previous generations of transmission equipment, such as the plesiochronous digital hierarchy (PDH), is the software configurability of equipment features and parameters and the capability to achieve this remotely. A small number of operators at a single location can then remotely configure and reconfigure the network elements throughout the network, setting up and clearing down customer connections (such as a 2 Mbit/s MegaStream Genus circuit) at the click of a mouse on a screen. They can also carry out a number of network management operations such as audits, inventory checks and software downloads, etc, all remotely from the management centre.

To achieve this manageability of the network, communications must be established and maintained from the management systems to every element in the network. The SDH standards include two embedded communications channels designed specifically for this purpose, contained in bytes D1 to D12 within the overhead area of the SDH frame. The same management channels are also used by the network elements to send their alarm and other event information back to the management systems.

The two embedded control or management channels in the SDH signal are called the DCCr management channel and DCCm management channel (data communications channel regenerator and multiplexer respectively). These are also collectively known as the embedded communications channels (ECCs).

A single DCCr management channel at 192 kbit/s is provided in bytes D1 to D3 as part of the SDH signal, irrespective of whether it is an STM-1, STM-4 or STM-16 signal. It is designed to carry management information to the regenerator elements of a line system.

A single DCCm management channel at 576 kbit/s is provided in bytes D4 to D12 as part of the SDH signal, irrespective of whether it is an STM-1, STM-4 or STM-16 signal. It is designed to carry management information to the multiplexer elements, or the terminal elements of a line system.

1.2 OSI and communications standards

The data communications with the network elements uses the 7-layer model and appropriate OSI (open systems interconnection) standards from the ISO (International Standards Organisation) as detailed in ITU SDH recommendations [1, 2].

In general, the communications use a connection-oriented layer 4 transport connection [3] (to ensure a reliable, guaranteed end-to-end delivery mechanism), a connectionless network (layer 3) protocol (CLNP) [4] and an HDLC-based (high-level data link control) layer 2 mechanism over each DCC. The complexity of the OSI 'Q' stack is illustrated in Fig 1 where the relevant ISO and ITU (ex-CCITT) standards are shown for each layer.

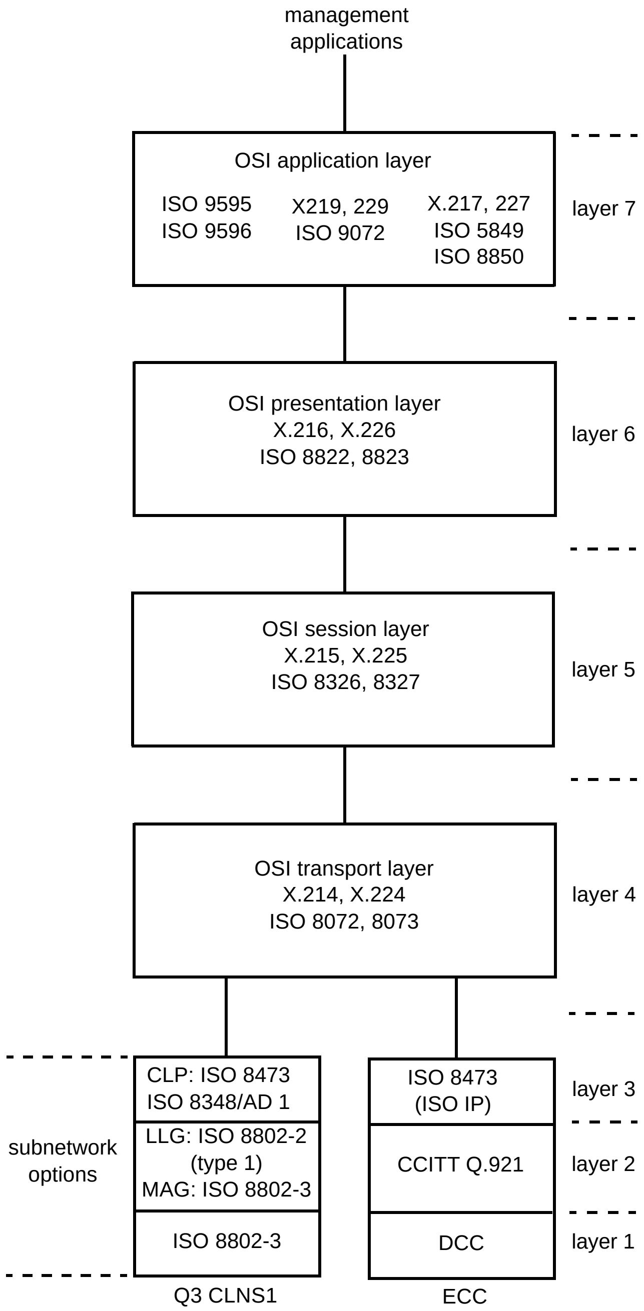

Fig 1 Element manager to network element DCC stack and LAN stack from ITU Rec G.784.

In accordance with the OSI layer 3 network level standard, each network element and element manager is assigned a unique network service access point (NSAP) address. All messages are labelled with their source and destination NSAPs to ensure delivery to the correct destination.

Alongside the OSI protocols the data communications network also uses transmission control protocol/Internet protocol (TCP/IP) where terminal access is required or where OSI protocols have not been implemented. The consequence of this is that there are often situations where NSAP and IP addresses co-exist on the same system. This mix of protocols is illustrated later in section 3.

2. BT SDH network architecture

2.1 SDH traffic architecture

The BT SDH network consists of two parts — the 'inland' network, and the 'global' network. Currently the inland network consists of equipment supplied by GPT/ Siemens, and the global network is dominated by equipment from Ericsson, although there is some equipment from ECI. These form a tiered traffic network as illustrated in Fig 2. The synchronous crossconnects (SXCs) are in tiers 0 and 1 and in the supercells to tier 1.

2.2 Description of the network

Tier 0 global network (Ericsson)

Tier 0 essentially consists of frontier stations and international transmission centres interlinked by the BT international backhaul network. Frontier stations deal with satellite and submarine cable entry into the UK, with Madley being an example of a satellite earth station and Lands End an example of a landing station for submarine cable systems. International traffic will either be required to terminate in the UK or to transit the UK. Crossconnects will normally be found at the frontier stations for testing, grooming and restoration purposes and these sites are interconnected and also connected back to the international transmission centres in London using STM-16 (2.5 Gbit/s) line systems.

Tier 1 and 2 inland network (GPT)

The BT UK inland SDH network primarily consists of three types of network elements — add/drop multiplexers, line systems and crossconnects. Together these form an integrated, hierarchically tiered traffic network, operating at a variety of SDH traffic levels. At present most of the customer access part of the network is served by STM-1 (155 Mbit/s) add/drop multiplexers. Tiers 2/3 consist of predominantly STM-4 (622 Mbit/s) add/drop multiplexers, with the long-distance tier 1 or trunk parts of the network consisting of large SDH crossconnects and STM-16 line systems. There are also various intermediate levels of the network, such as the supercells or diversity supercells, which consist of STM-4 (622 Mbit/s) add/drop multiplexers and mini-crossconnects.

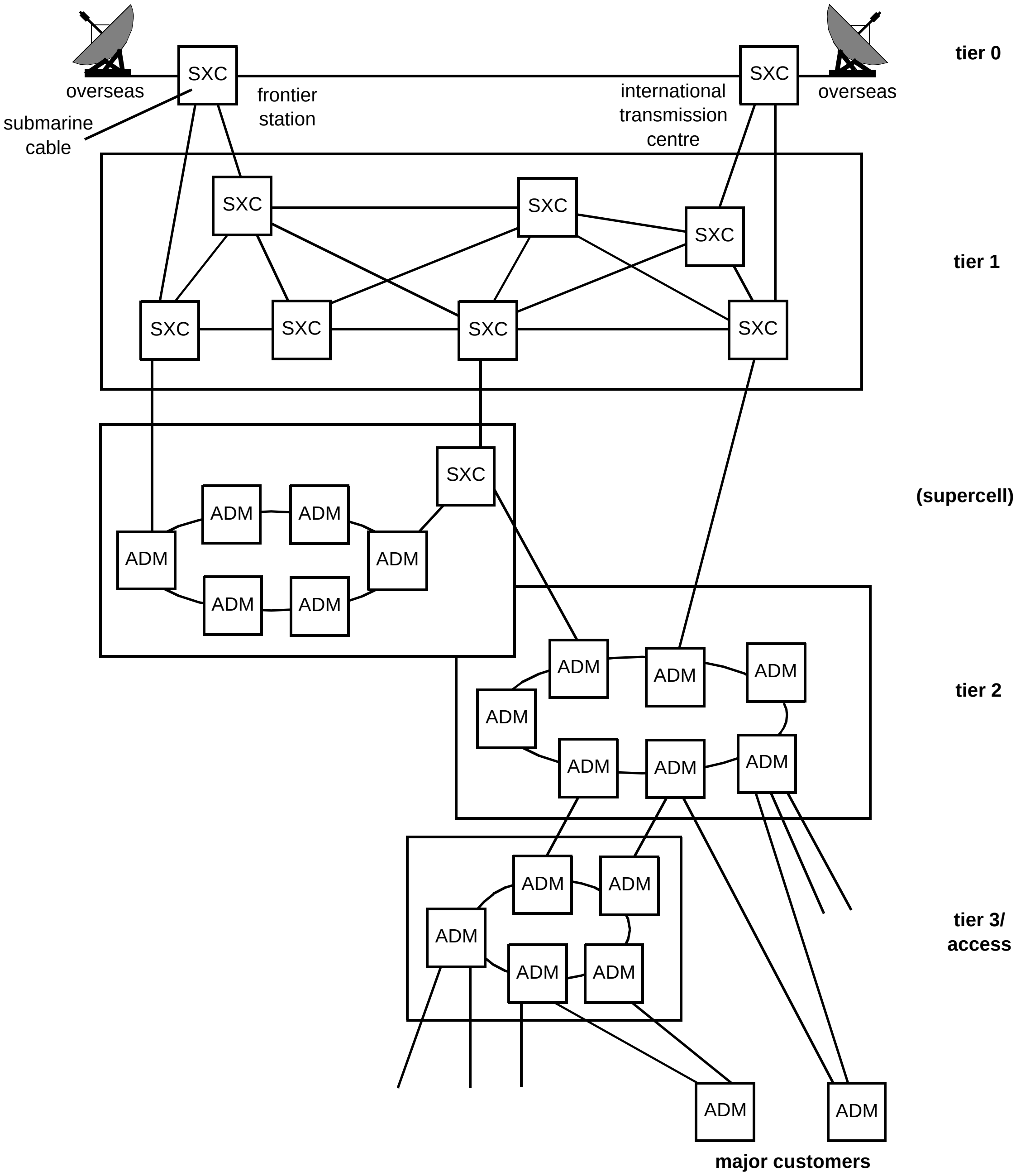

Fig 2 SDH tiered network.

The initial SDH network (I-SDH) was geographically spread in and around London to serve the City; the network is now expanding throughout the country.

Most major towns and cities in the UK now have at least some tier 2 customer access networks, a large or a mini SDH crossconnect and multiple STM-16 line system links as part of the main tier 1 network.

At present, the network consists of approximately 3100 add/drop multiplexers, 1200 line system elements and 75 large crossconnects each with a switching capacity up to 80 Gbit/s, some of which are shortly to be expanded to 160 Gbit/s. This network is growing at a rapid pace.

2.3 SDH management architecture

The implementation of the SDH network follows the principles of a Telecommunications Management Network (TMN) which are described in ITU-T Recommendation M.3010 [5]. The TMN principle is illustrated in Fig 3. All the SDH network elements are managed remotely by various types of operations systems (OSs) and consequently all rely heavily on the SDH data communications network to remain manageable.

The OSI-based management hierarchy is illustrated in Fig 4 At the bottom layer are the SDH network elements (NEs) (add/drop multiplexers, line systems, and

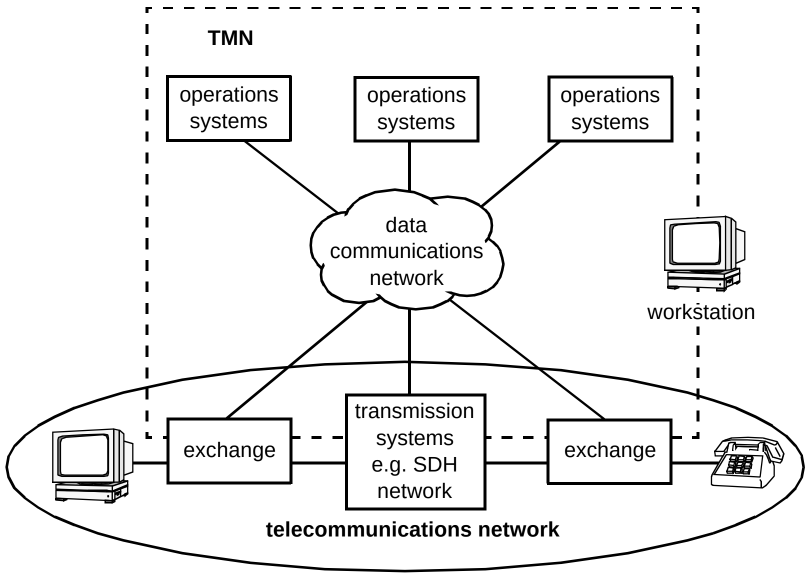

Fig 3 General relationship of a TMN to a telecommunications network.

crossconnects) which are managed by their associated element managers (EMs). A single EM may manage many NEs. In the SDH network a single EM usually manages 50 or more multiplexers or line systems. The EM for the crossconnect may only manage one such equipment due to its large size. Above the EMs is the network management level. At the present time the service and business management layers are less developed both within the international standards and within the BT network. The network management layer contains numerous OSs performing various functions such as the setting up of traffic paths across the network and fault management.

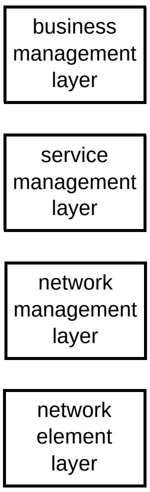

Fig 4 Conceptual management hierarchy.

2.4 Management of the network

The tier 0 and tier 1/2 networks are managed individually and have separate data communications networks (DCNs), but they are managed in a similar way.

Each network can be managed from either of two BT operations centres, called network operations units (NOUs), which provide protection against catastrophic failure of a NOU — if one fails the other can take over.

The NOUs manage the network through a number of regional ‘lights out’ management centres and centralised management systems at two main computer centres . At present there are about 60 such regional centres located throughout the country, from Exeter to Edinburgh. These centres are linked to the NOUs through the two main computer centres which house a number of the higher level management systems and also form the hub of a resilient, high-capacity wide area communications network. Figure 5 illustrates the interconnection of the main computer centres, NOUs and regional centres.

Two layers of network management are implemented — the element management layer and the network layer. The element management layer consists of the element management systems which each directly manage a part of the network, and the network level management systems manage the whole of the network, working through the various element managers.

The element management layer at present consists of approximately 160 element management systems, distributed around the country. Each of these regional centres is used to host up to 8 element management systems to manage the SDH network elements within that region of the network. These regional centres range in size, as determined by the numbers of elements they are required to manage.

The network level management systems are accessed and used by the network operations staff at the NOUs, either through the high-level management systems or accessed directly using X-Windows sessions.

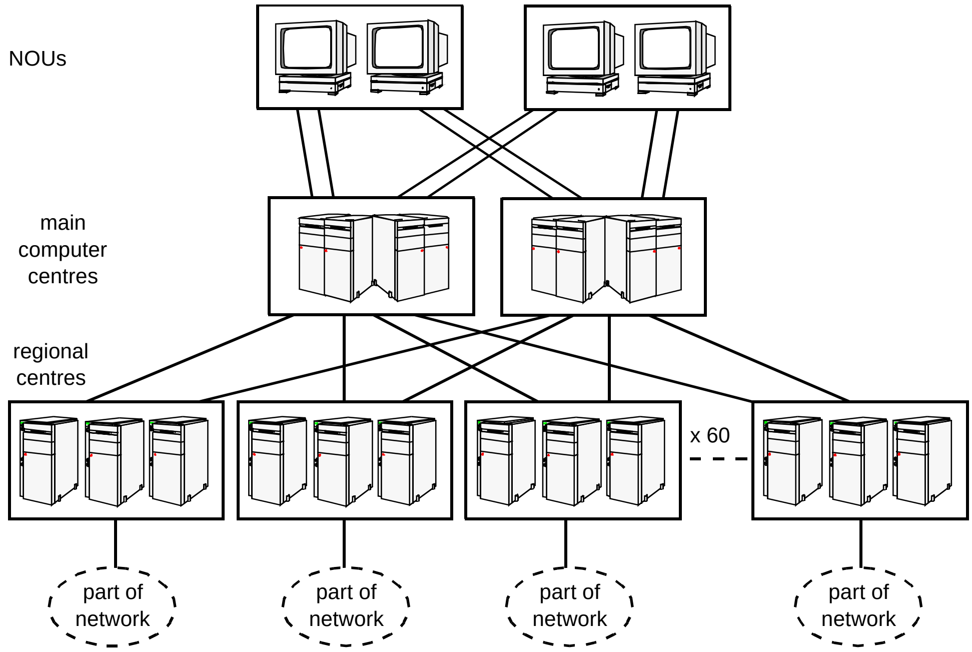

Fig 5 Interconnection of management sites in the network.

The actual network level management systems for the core network (tiers 1 and 2) are located at the main commuter centres at London and Manchester. These network level management systems have been developed by BT Laboratories (BTL) to allow the operators to see and manage all of the network as a single entity, even though it is actually managed at the element management layer as a number of smaller interconnected, but discrete regional networks. They also interoperate with other BT systems, for global fault analysis, network planning and customer service ordering.

Communications between the various sites of the network are maintained by a network of standard commercial Ethernet routers and a combination of KiloStream and MegaStream point-to-point links. Together these form a highly resilient country-wide communications network, capable of maintaining full network management, in the event of any single equipment or link failure or site loss.

3. The management communications implementation

Management communications traverse the network by various means depending upon the portion of the network concerned. Thus the interface between EM and NE is different from that between EM and the network management layer (NML). Communications between the NEs themselves can be via embedded channels within the SDH frame structure.

The DCN paths that the management communications take through the network is illustrated in Fig 6.

To provide the management services required, the DCN has to support the message data volumes generated as a result of the following network activity:

- configuration changes,
- alarms and status change notifications,
- error performance reports,
- software downloads,
- data audits.

In addition the DCN has to support various control-type messages such as those required by the routeing protocols being operated, and heartbeat-type messages between the various management systems and the network elements.

3.1 EM to NE communications

For the add/drop multiplexers or line systems to element manager, the communications protocols are based on the OSI recommendations as specified by the International Standards Organisation (ISO). Communications to the network elements generally utilise capacity provided within the SDH overhead specifically for management communications.

For add/drop multiplexers and line systems, the element managers communicate via Qx interfaces to a gateway network element (GNE). A GNE provides the entry point from the element manager into the SDH network elements that it is managing. The element managers and the GNEs are connected via a 10base5 Ethernet for the core network and 10baseT via hubs for the GN network.

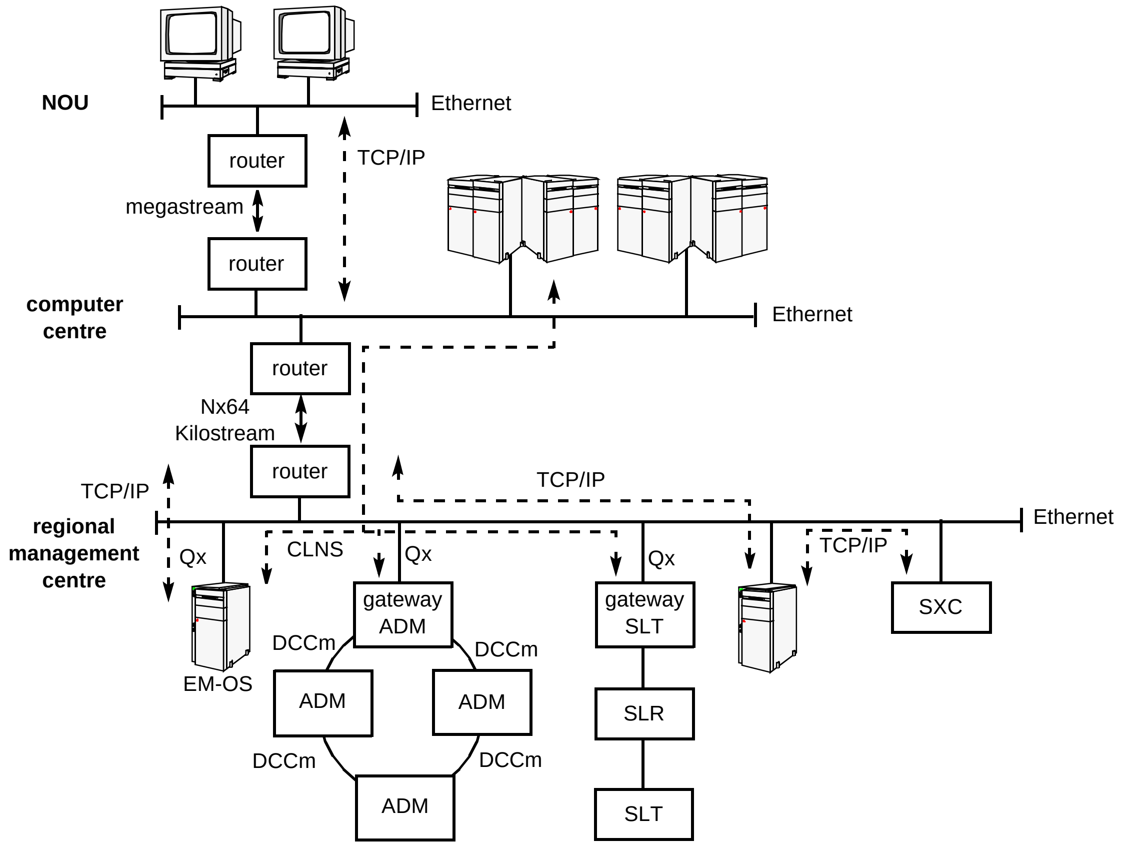

Fig 6 DCN paths providing management communications activities.

The crossconnect is closely coupled to its element manager on a one-to-one basis with a dedicated link; this communication does not traverse the DCN.

3.2 NML to EM communications

All the element managers are connected to an Ethernet. The element managers communicate to the network management layer via the router network. The routers are interconnected by point-to-point links which are $N \times 64$ KiloStream links provided by BT. The capacity of these links depends upon the size of the site. This traffic is generally TCP/IP based, utilising TCP at layer 4 to provide a robust, connection-oriented protocol.

The network administration computer centres (NACCs) and NOUs are connected via routers with MegaStream point-to-point links.

3.3 Resilience

The SDH network carries very large volumes of traffic and supports numerous customers. It is therefore essential that the communications network is highly reliable. Any loss of communications to a network element or manager can severely impact on the visibility of the service being offered although existing traffic would not be affected.

When designing the DCN the principle was to provide at least two diverse paths to any particular site, management system or network element. These are achieved by means of two separate point-to-point links and routers.

There are two GNEs between each element manager and add/drop multiplexer or line system. In certain instances, in order to provide resilience, a point-to-point bridged link is provided when there is no convenient alternative path through the SDH network to the secondary GNE.

At the network element level, due to the highly meshed nature of the network, there are generally numerous diverse routes that management messages can take. The decision as to which way to forward, or route, messages is performed within the network elements by the routeing algorithm being operated. This may be by the use of either static methods such as routeing tables, or dynamically by the use of a link state routeing algorithm such as ISIS [6].

4. Intermanagement communications — the router network

Despite the TMN promise of a self-contained DCN, an external data network is still needed. The reason for this is that the BT SDH network is managed from centralised sites which are not collocated with SDH transmission sites. Therefore a method is required to carry the management traffic between the SDH equipment and the management sites. Figure 7 shows the data communications domains.

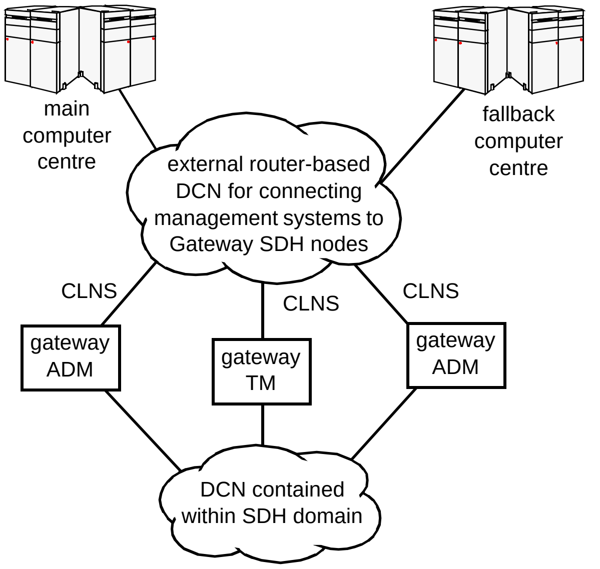

Fig 7 The data communications domains.

A router network is used to implement this requirement, with KiloStream and MegaStream links being used to connect the sites. The routers extend the connectionless network service (CLNS) domain across the KiloStream links so that the management traffic can reach the gateway network elements.

The integrity of the DCN is critically important to maintain manageability of the SDH network. Loss of manageability due to router failure is avoided by using two routers at each node — if one fails then the management traffic is diverted to the other (see Fig 8). Loss of manageability due to KiloStream failure is avoided by using a minimum of two KiloStream links, with separate routeing, to each node.

In addition to the CLNS DCN, a TCP/IP network is required to handle non-CLNS communication. This TCP/IP traffic is for:

- operator work-place to management system communication — although the management systems communicate to the SDH network elements using CLNP, there is also a need to communicate between the management system and the operators' work-places,
- TCP/IP-based management systems — some of the management systems are based on TCP/IP instead of CLNS (primarily the early 4/1 crossconnects),
- simpler management systems — some of the supporting equipment for the SDH network, e.g. synchronisation units, rely on basic management communications, typically based on an RS-232 connection, which can be encapsulated in TCP/IP and sent back to a central site,
- management of the DCN network — because of the critical nature of the DCN, it is important that the equipment from which it is made up is also managed; SNMP-based (simple network management protocol)

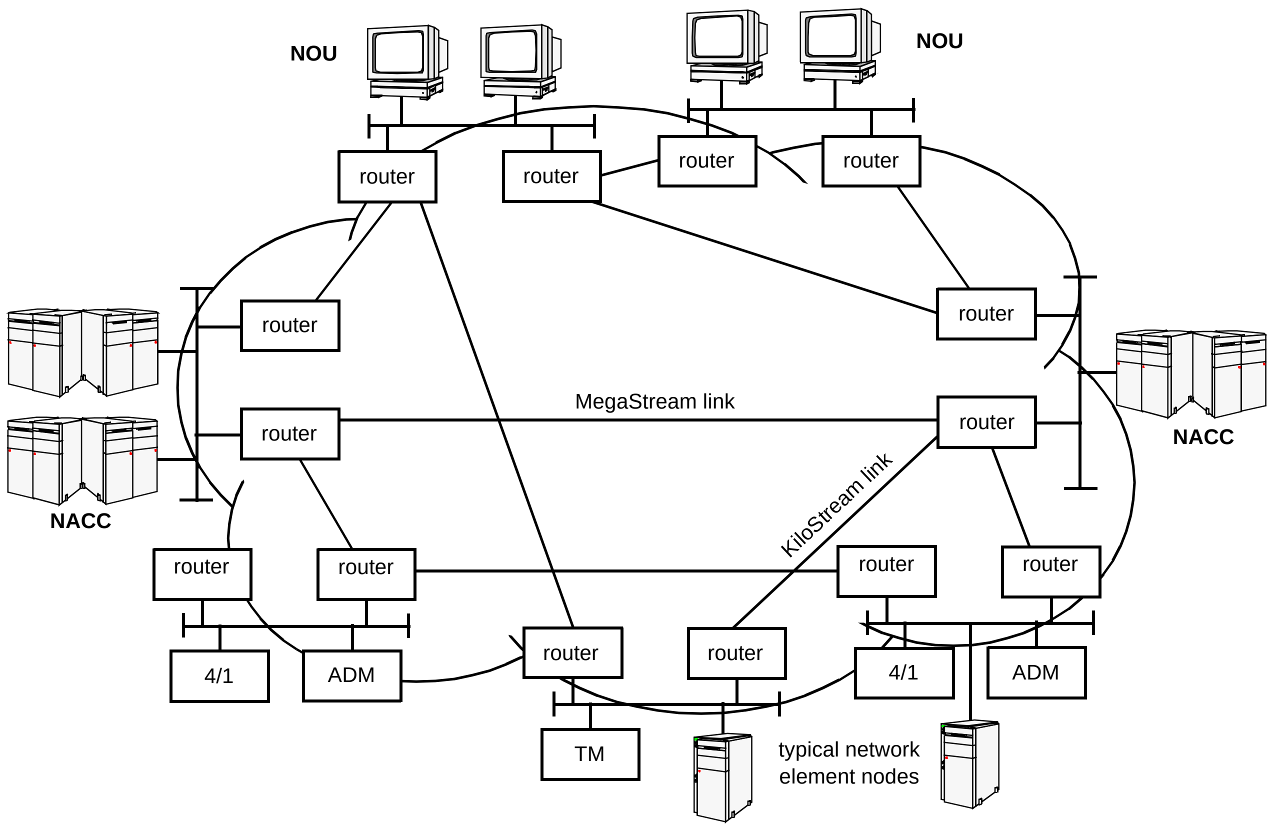

Fig 8 The protected data communications network.

management systems (e.g. SunNET Manager) are used to manage the routers, Ethernet switches and Ethernet hubs, and also to monitor the computer equipment (this SNMP management traffic is carried over IP).

As the 4/1 crossconnects are distributed within the SDH network, the TCP/IP network has to be extended to these sites (see Fig 9). Fortunately, as most 4/1 sites are also gateway nodes for the SDH line systems, the routers can therefore carry both TCP/IP and connectionless network protocol (CLNP — used by CLNS). Likewise, the servers for the distributed systems are normally located at the major SDH equipment nodes which also contain routers for the 4/1s and the gateway network elements.

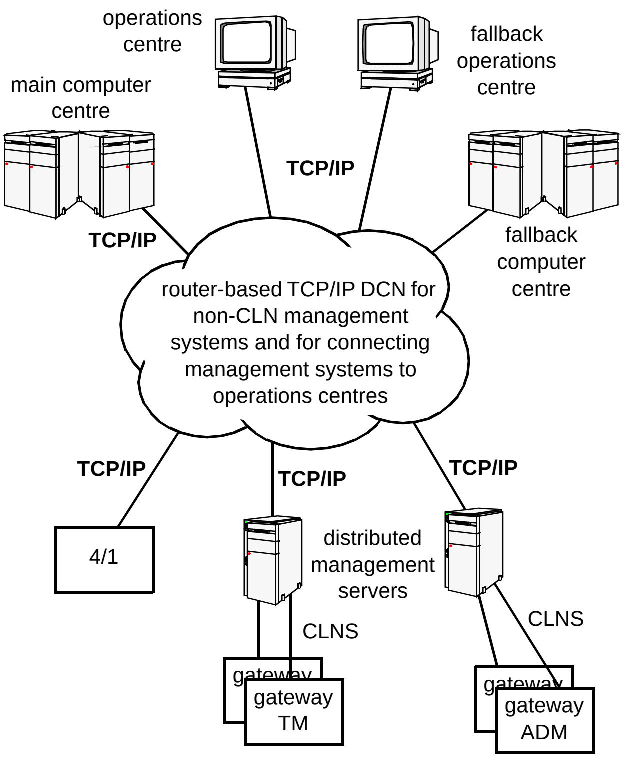

Fig 9 The use of TCP/IP.

5. Communications between element managers and network elements

5.1 Communications with line systems and multiplexers

As discussed previously, the BT core network is managed at the element level as a number of discret e regional networks, with each region or 'sub-network' managed from a regional management centre where the EM-OS and operational and maintenance terminal (OAMT) element management systems are located. The add/drop multiplexers and line system terminals and repeaters in that sub-network are managed by the element managers at that centre, as the crossconnects or SXCs are managed by the OAMTs at the centre.

The EM-OSs and OAMTs are based on standard commercial Unix workstations with Ethernet (10base5) connectivity. The Ethernet routers being used to provide the NACC-NOU regional management centre connectivity also have Ethernet ports, as do the gateway network elements, which are also located at the regional management centres. Ethernet is then used as the primary communications medium within the centres, providing 10-Mbit/s connectivity between all of the equipment at each centre. A simple regional management centre example is illustrated in Fig 10, which shows a sub-network managed from that centre. Actual sub-networks controlled from a regional management centre can contain up to 300 or more elements.

Figure 10 shows an example regional management centre site, with one SXC crossconnect and its local OAMT element manager, two EM-OS element managers, together managing two small rings of synchronous multiplexer add/drop (SMA) and a three-element line system. In this case, the network has been designed with traffic considerations in mind and as such there is only a DCC or embedded control channel access into one end of the line system. To avoid this configuration having non-resilient data communications, a bridge link is shown from the centre Ethernet to the other end of the line system.

The communication paths between the EM-OS and associated network elements (SMAs, SLTs and SLRs — synchronous line terminal or repeater) use the embedded communications channels available within that sub-network wherever possible. In this example, the EM-OS controlling ring 1 would normally route its messages directly into ring 1 via gateway 1, as would the EM-OS controlling ring 2 normally send its messages directly into gateway 2. Messages bound for the line system would normally be routed via gateway 2. Once inside a ring or line system, those messages would then route around the ring or down the line systems until they got to their destination. Messages returning from the network elements to the EM-OSs would follow the reverse route.

Under fault or other abnormal conditions, however, messages will reroute as necessary, so that messages for ring 1 elements could be sent via gateway 2, or messages for the line system could be sent over the bridge link, for example.

This routeing and rerouteing of messages is automatic and can be highly resilient, delivering messages to elements even while several faults may occur in the network. There are a number of routeing algorithms that can be used to control this routeing of messages. These are explained in more detail below.

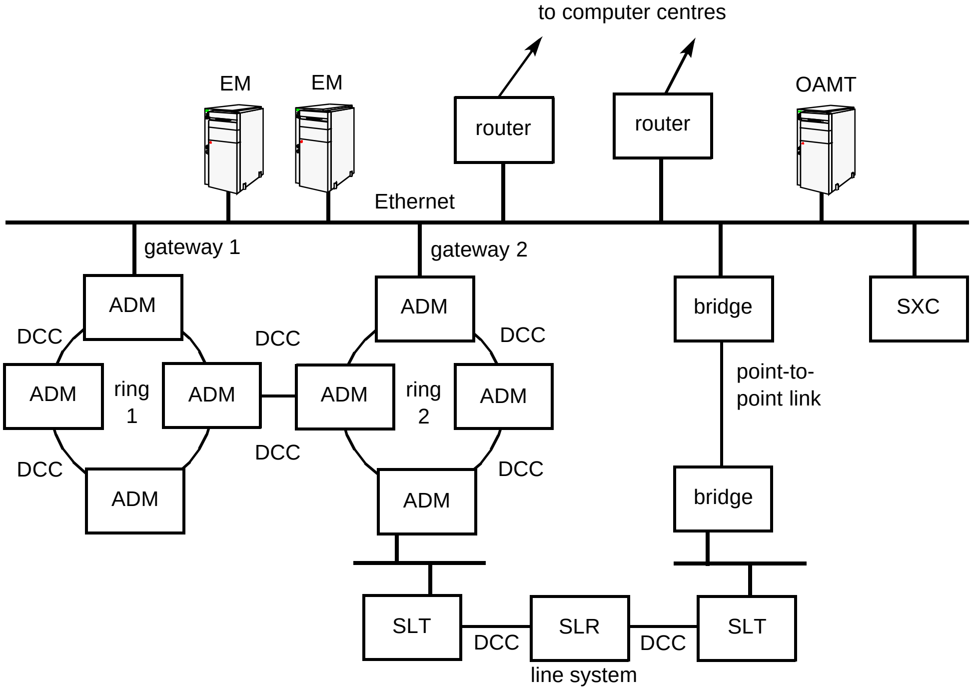

Fig 10 Data communications schematic for an example tier 2 regional management centre.

Each crossconnect is normally managed centrally, but can also be managed through a local manager for secondary access. The centralised managers use the router network to communicate with the crossconnect, the local managers are directly connected by Ethernet.

The overall communications path between the operators at the NOUs and the network elements is over the router network (and its KiloStreams and MegaStreams) to the element managers at the regional management centres and then over the DCCs and possibly bridges to the SMA and SLA network elements themselves.

5.2 Communications with crossconnects

Each of the first generation crossconnects has a dedicated, collocated, element manager. They use proprietary management communications carried in TCP/IP across a private point-to-point Ethernet link for EM-NE communications. This is so that other management traffic at a node does not interfere with the management of the crossconnect. Communication from the crossconnect element manager to the network management layer and/or the operator terminals is also TCP/IP based and is similar to the EM-NML and EM-operator terminal communications of the other element managers.

Ericsson second generation crossconnects are centrally managed and use ITU Q3-based management. Communications with the EM, which is based at the main computer centre, uses CLNS in a similar way to the line systems. However, as these crossconnects are typically located at router nodes their management traffic does not need to be carried in the DCC — it can go straight into the router CLNS network.

5.3 Resilience of the data communications below the element managers

Each sub-network has been designed for resilience, so that the matrix of DCCs used provides at least two physically independent paths to every network element. Should a fault, such as a fibre break, occur on any connection, then the management traffic would transparently and independently switch from using the failed DCC to one that is operational, immediately bypassing the fault and maintaining communications with all network elements.

Where the traffic network does not provide suitable DCC management channels, or it is decided not to use some of the available DCC data communications channels, then a direct or bridged Ethernet connection between elements can also be used. For example, a direct Ethernet link to carry the data communications around a patch panel is often implemented, so that the traffic connections can be re-patched as necessary, without concern for the DCC management communications channels embedded in those traffic connections. Bridged Ethernet connections use standard commercial Ethernet bridges and a point-to-point service, such as KiloStream or MegaStream. Such bridged data communications connections are usually only implemented to provide a secondary path for resilience purposes, if no suitable DCC path exists.

This dual path principle is illustrated in the network shown in Fig 10. The communications network design is such that there are always at least two independent routes from every EM-OS to each and every element under their control. If the first fails for any reason, then the second will be used, maintaining communications and management of all elements.

5.4 Routeing of data communications below the element managers

The traffic network in each region or sub-network managed from every transmission station (TRS) will be unique and will hence provide a unique matrix of DCC connections between its elements. When messages are sent out from EM-OS destined towards an element or from an element destined towards its EM-OS, then that message could theoretically take any one of a number of potential paths to reach its destination. As the messages pass from element to element over each individual DCC channel, then every element through which it passes must make a decision as to where to send that message next.

In cases of an element which has only two connections (e.g. the SLR 1 in Fig 10), then it is reasonable to assume that, if it came in on one of those ports, it should continue its journey by going out on the other. In elements with more than two DCC or Ethernet connections from which to choose, the selection of which port to use may be more complicated. Some add/drop multiplexers can have over 40 such DCC channels and multiple Ethernet destinations from which to select. The decision of which path or route to take next is called ‘routeing’. An introduction to routeing and routeing protocols can be found in King [7]; advanced IP routeing protocols are discussed in Whalley et al [8] and O’Neill et al [9]

The purpose of routeing is to deliver all messages over their shortest path from their source to their destination, to ensure the fastest delivery time and hence the fastest management and response times possible. The routeing must also automatically accommodate problems in the network, such as fibre breaks or power failures to particular elements. The neighbouring elements must detect these problems and dynamically select the best remaining route, over which they can transmit any messages which would have been sent over the failed route.

At present, routeing decisions are mainly based on one of two routeing methods, either a predefined, factory-set routeing algorithm or a routeing table. The default algorithm is only used if that element is deployed in a simple part of the network where complex routeing decisions will not be required, such as element SLR 1 in Fig 10.

In more complex sub-networks, such as where more complex routeing decisions must be made, then routeing tables are deployed, defining up to five potential routes or next hops for each destination, in priority order. This then takes over the routeing decisions in that element. The ‘quasi dynamic’ routeing table algorithm will use the first or primary route to all destinations under normal conditions, dynamically switching to the second route only if the first is not available due to a fault. The algorithm will also detect looping messages and try the next available route for that destination, as listed in its table.

If that second route is also unavailable or loops again, then the third route will be selected, and so on, until all five potential routes to that destination have been tried. Messages to other destinations will not be affected, unless they are detected to be looping or are off their primary route. The routeing table algorithm will automatically revert to the highest priority route available to any particular destination, should a previously failed route become available again. These routeing tables are pre-generated, fixed tables, specifically for the sub-network in which they will be used.

The next version of firmware that will be downloaded into the majority of the UK core network will implement an enhanced routeing protocol called ISIS (intermediate system to intermediate system) (ISO 10598). This will replace the routeing table algorithm and its routeing tables, to provide a fully dynamic network. Using this routeing protocol, each element will dynamically learn the current network topology ‘on the fly’, automatically creating the most appropriate routeing tables for that network at that instant in time. This will offer significant advantages and less maintenance over the present type of fixed routeing tables.

5.5 Future evolution

Although of considerable size already and expanding fast, the BT SDH network is still in its infancy compared to the existing PDH transmission network. The SDH network is expected to grow considerably over the next few years, with predictions of over 20 000 elements within a few years.

Much of this growth will come from increased numbers of the existing types of network elements in the UK core network, but new products with different or improved characteristics will also be deployed. Growth in the global network will be dependent on sub-sea cable systems and satellite communications which are both considerably more expensive to install and more complex to organise than any system in the inland network.

One of the new products for the inland network will be a GPT/Siemens sub-STM-1 system of which there are currently two variants, 8×2 Mbit/s or 14×2 Mbit/s port sizes, both planned for large-scale use on customer premises, with dual links into the network.

One interesting evolution aspect of the network will be to feed the inter-site point-to-point links (currently KiloStreams and MegaStreams) back over the SDH network itself as paths or possibly protected paths. This will remove the dependency of the SDH network on the older PDH network, thereby giving the operators full visibility of all their own data communications networks. This could be used for all NACC/NOU/TRS links, as well as bridge links. With careful planning the dual path protection and the increased reliability of SDH equipment should provide an extremely reliable data communications network to serve SDH well into the new millennium.

This large-scale deployment of customer premises equipment is expected to be a growth area which will have significant impact on the SDH data communications requirements, especially on the loading of the DCC.

6. Conclusions

SDH is the first of a new generation of technologies which has in-built complex management capability and gives rise to the need for a resilient data communications network. This paper has shown many of the factors which influence the design of SDH data communications networks and how this has been put into practice.

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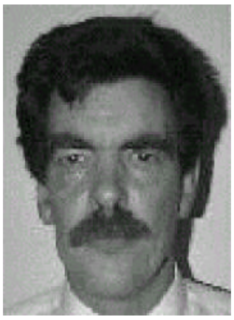

Roger Cole joined BT in 1969 initially working on transmission maintenance and provisioning duties. He was subsequently involved in the approval of first generation 30-channel PCM systems. He played a key role in the specification and approval of NICAM digital sound programme transmission equipment, and was involved in the specification and evaluation of BT’s KiloStream ACE and its associated management system (RENACE). More recently he has played a key role in formulating BT’s requirements for a managed transmission network based on synchronous digital hierarchy and currently is focused on data communications and volumetrics for both SDH suppliers.

Andy Crow graduated at City University with a degree in Electronic Engineering. He joined BT in 1980 initially working on the evaluation and design approval of PDH line systems. He later moved on to PDH higher order multiplex and automatic service protection systems. He played a key role in the introduction of SDH into the BT network, concentrating on BT’s philosophy for event and performance management.

In his current role he is responsible for SDH management communications.

David Davenport graduated from Manchester University in 1982 with a degree in Electrical and Electronic Engineering. Following his degree, he worked for Marconi Radar Systems Ltd, initially as an electronics development engineer and later as a firmware engineer. He later worked for Data Sciences Ltd and then Digital Equipment Ltd. He presently works for GPT Ltd, managing the design of the data communications and management aspects of the GPT/Siemens-supplied BT SDH network.

Chris Spencer-Smith graduated with an honours degree in Mathematics from Southampton University and started work as a software engineer on System X with GEC Telecommunications. After working for Marconi and Rolls Royce in the Radar and avionics industries, he joined Ericsson as a Principal Systems Engineer where he is responsible for the design of the SDH management and data communications networks. He is also actively involved with the specification and introduction of BT’s Home Highway service.